Apéry $\zeta(3)$: the Zeilberger certificate we need for the last Lean sorry

Zinan — internal note for Xiang

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1 Setup (definitions)

Working in $\mathbb{Q}$. For $n, k \in \mathbb{N}$ with $k \le n$ define

$$\begin{aligned}
P(n,k) &:= \binom{n}{k}^2 \binom{n+k}{k}^2 \\
B(n,k) &:= 4(2n+1)\big(k(2k+1) - (2n+1)^2\big) \cdot P(n,k) \\
H_3(n) &:= \sum_{m=1}^{n} \frac{1}{m^3} \\
e(n,k) &:= \sum_{m=1}^{k} \frac{(-1)^{m-1}}{2m^3 \binom{n}{m}\binom{n+m}{m}} \\
c(n,k) &:= H_3(n) + e(n,k)
\end{aligned}$$

The Apéry sequences are

$$a_n := \sum_{k=0}^{n} P(n,k), \qquad b_n := \sum_{k=0}^{n} P(n,k)\, c(n,k).$$

Both are rational; a_n is in fact a positive integer, and $2\,[1,2,\ldots,n]^3\, b_n \in \mathbb{Z}$ (key integrality for the irrationality proof).

Let

$$F_1(u)(n) := (n+1)^3\, u_{n+1} \;-\; (2n+1)(17n^2+17n+5)\, u_n \;+\; n^3\, u_{n-1}.$$

(Note: $(2n+1)(17n^2+17n+5) = 34n^3 + 51n^2 + 27n + 5$, as in vdPoorten.)

2 What is proved (Lean, `AperySequences.lean`)

F1 for a_n (proved): $F_1(a)(n) = 0$ for $n \ge 1$. Standard Zeilberger telescope with witness $B(n,k) \cdot P(n,k)$, formalized in Lean as `aperyA_recurrence`.

Decomposition (proved): $b_n = H_3(n)\, a_n + d_n$ where $d_n := \sum_{k=0}^{n} P(n,k)\, e(n,k)$. Formalized as `aperyB_eq_decomp`.

Harmonic correction (proved): $F_1(H_3 \cdot a)(n) = a_{n+1} - a_{n-1}$ for $n \ge 1$. Formalized as `aperyHA_recurrence`. (Derivation: shift identities $H_3(n+1) = H_3(n) + 1/(n+1)^3$ and $H_3(n-1) = H_3(n) - 1/n^3$ plus $F_1(a) = 0$.)

Target (sorry'd, this document's subject):

$$F_1(d)(n) \;=\; a_{n-1} - a_{n+1} \qquad (n \geq 1). \tag{1}$$

Equivalently, $F_1(b)(n) = 0$: since F_1 is linear,

$$F_1(b) = F_1(H_3 \cdot a) + F_1(d) = (a_{n+1} - a_{n-1}) + F_1(d),$$

so (1) holds iff $F_1(b) = 0$. The file states both: `aperyD_recurrence` (sorry) and `aperyB_recurrence` (currently derived from `aperyD_recurrence` via `linarith`).

3 The exact question for external research / ChatGPT

Question. Find a (rational in n, k) witness $W(n,k)$, defined for $n \geq 1$ and $k \in \mathbb{Z}$, with $W(n,-1) = 0$ and $W(n,k) = 0$ for k sufficiently large, such that

$$(n{+}1)^3 P(n{+}1,k)\, c(n{+}1,k) \;-\; (2n{+}1)(17n^2{+}17n{+}5)\, P(n,k)\, c(n,k) \;+\; n^3\, P(n{-}1,k)\, c(n{-}1,k) \;=\; W(n,k) {-} W(n,k \tag{2}$$

Summing (2) over $k \geq 0$ telescopes to $W(n,\infty) - W(n,-1) = 0$, giving $F_1(b)(n) = 0$.

4 Answer from vdPoorten (1979), §8 p. 201

I had skipped this; the explicit certificate IS in the paper. After expanding (2) as

$$(B_{n,k} - B_{n,k-1})\, c_{n,k} + (n+1)^3 P(n+1,k)\big(c_{n+1,k} - c_{n,k}\big) - n^3 P(n-1,k)\big(c_{n,k} - c_{n-1,k}\big)$$

(the standard decomposition combining the a_n Zeilberger telescope with the c-shifts) and *massive reorganization*, the witness is

$$\boxed{W(n,k) \;=\; B(n,k)\, c(n,k) \;+\; \frac{5(2n+1)(-1)^{k-1}\, k}{n(n+1)} \binom{n}{k}\binom{n+k}{k}.} \tag{3}$$

That is: the standard $B \cdot c$ piece (the "naive" telescope from the a_n certificate carried through c) plus a simple rational correction.

5 Numerical verification of the witness (passed)

I wrote a Python check (`/tmp/verify_witness.py`) using exact rationals (`fractions.Fraction`) over $\mathbb{Q}$. For each $n \in \{1,2,3,4\}$ and each $k \in \{-1,0,1,\ldots,n+2\}$ it compared

$$L(n,k) := (n{+}1)^3 P(n{+}1,k) c(n{+}1,k) - (2n{+}1)(17n^2{+}17n{+}5) P(n,k) c(n,k) + n^3 P(n{-}1,k) c(n{-}1,k)$$

with $W(n,k) - W(n,k-1)$ for W from (3). **All 24 pointwise checks matched exactly.** Also $\sum_k L(n,k) = 0 = F_1(b)(n)$ for $n \in \{1,\ldots,5\}$.

6 Boundary behavior (both zero)

- $W(n,-1) = 0$: the main piece $B(n,-1) \cdot c(n,-1) = 0$ because $B(n,-1) = 0$ (by the Zeilberger convention: $P(n,k) = 0$ for $k < 0$), and the correction is 0 because of the factor $k = -1 \cdot \ldots$ wait, the factor is $k = -1 \neq 0$. Actually: $\binom{n}{-1} = 0$, killing the correction. Good.
- For $k \geq n+2$: $B(n,k) = 0$ (since $P(n,k) = 0$ for $k > n$) and $\binom{n}{k} = 0$, so $W(n,k) = 0$. Hence $\sum_{k=0}^{N}\big(W(n,k) - W(n,k-1)\big) = W(n,N) - W(n,-1) = 0 - 0 = 0$ for $N \geq n+1$.

7 What this means for the Lean formalization

Previously the sorry was "prove vdPoorten's massive reorganization identity." With (3) verified numerically, the remaining Lean work is *mechanical*:

1. State (2) as a `have` lemma (pointwise polynomial-in-$\mathbb{Q}$ identity). Prove by `field_simp; ring` after unfolding B, P, $c = H_3{+}e$, and expanding the e-differences via `aperyE_diff_right_closed` (already proved).

2. Sum both sides over $k \in$ `range (n+2)`. RHS telescopes via `Finset.sum_range_succ` and `sum_sub_distrib`; the $W(n, n+1)$ endpoint vanishes by $P(n, n+1) = 0$.

3. Recognize LHS of (2) summed as $F_1(b)(n) = 0$, hence `aperyB_recurrence` (the genuinely-proved direction, not the current circular derivation).

4. `aperyD_recurrence` then follows in one `linarith` from `aperyB_eq_decomp`, `aperyHA_recurrence`, and `aperyB_recurrence`.

Estimated effort: 100–200 lines of Lean. Much smaller risk now that the witness is verified.

8 Follow-up: Section 9 gives the $\zeta(2)$ analogue for free

vdPoorten §9 notes that for $b'_n = \sum \binom{n}{k}^2 \binom{n+k}{k} c'_{n,k}$ (the $\zeta(2)$ case), the analogous witness is

$$B'(n,k) = \big(k^2{+}3(2n{+}1)k{-}11n^2{-}9n{-}2\big)\binom{n}{k}^2\binom{n+k}{k}, \quad A'(n,k) = B'(n,k)c'_{n,k}{+}3(-1)^{n+k-1}\frac{(n-1)!}{(k-1)!}.$$

So once we have the $\zeta(3)$ certificate formalized, the $\zeta(2)$ version costs only substitution and re-verification.